LIFELINE - Approach Note (Iteration 1), Group 40

Members: Hassan Tariq (24i-2555); Noah Faraz (24i-2585); Diya Hurmat (24i-0094). Group seed s = 40, case_id 40-00 .. 40-49. Every number comes from **generator/main.py**: each (case, module) pair has its own random stream seeded with the string "40-k-module", so the file is reproducible and changing one module never changes another. All sizes are drawn uniformly from the tier ranges of Section 3 (tier X uses tier L, except where a trap overrides a size).

1. From story to numbers

Case study	Main choices (sizes, distributions, shapes)	Story sentence that justifies them
1 Delta Flood	**Network:** each village joins one of the 3 previous villages, extra roads join villages 2-5 apart; average degree 2.3-3.0; 5-40 min per road. **Penalties:** 45% of locations flooded (2, 4 or 6), the rest 0 or 1. **Waves:** 3-5. **Crates:** 62% food/water.	"A flat delta of small villages joined by raised roads. Roads go under water in waves. Food and water matter most."
2 Mountain Earthquake	**Network:** random tree plus few extra roads, degree 2.0-2.5, 10-60 min per road. **Crates:** 52% medicine/tools. **Alarms:** 60% fall within 3 ticks after an aftershock tick; priority weights favour 6-10 (mean 6.6-7.2 in our cases).	"A valley town with a few roads and almost no alternatives ... medicine and tools. Bursts of urgent calls follow each aftershock."
3 Coastal Cyclone	**Network:** port hub: half of the locations attach to a few hub nodes; 32% shelters; 3-5 waves. **Windows:** 5-60 min, starting within 40 min of a surge peak (one peak per 40 windows). **Log:** 12% of symbols replaced by random noise, at least 32 symbols used.	"A port city and fishing villages. Storm surge closes roads in waves. Rescue windows are short and overlap. The radio is full of noise."
4 Urban Flash Flood	**Network:** street grid with diagonals, degree 3.6-4.6, 1-15 min per road. **Alarms:** 1.55-1.75 per tick on tiers M/L. **Crates:** 14% rare equipment.	"A dense city with many alternative routes. Calls arrive faster than anyone can answer them. Rare equipment is available."
5 Landslide Corridor	**Network:** one highway through 60% of the locations starting at the HQ; side valleys hang off it; no location has more than 4 roads; degree 2.0-2.3. **Waves:** 3-5; wave 1 always contains a road whose closure cuts a target off. **Windows:** long clearing jobs, 60-480 min.	"A single mountain highway along a gorge. A collapse cuts off whole communities."
6 Outbreak Camp	**Crates:** 62% medicine/water; 55% expire between hour 24 and 300. **Log:** 40% of messages are one beacon letter repeated 6-14 times; that letter is at least 30% of the log. Compact camp paths, 1-10 min per road.	"Medicine and water count and supplies spoil. Radio beacons repeat the same messages."

Common choices. Crate value = category range + 2 x weight - 20 (clipped to 10-100), so heavy crates are worth a little more and the knapsack is not trivial. Capacity is 30-60% (S), 20-40% (M) or 10-25% (L, X) of the total weight. The radio log is built from story words (FLOOD, BOAT, QUAKE, ...) and numbers, cut into cells of at most 10,000 symbols. Alarm ticks start at 0 and the last tick is chosen from the target rate (1.2-1.8 per tick on M, L and X, with at least 6 priorities including 1). Sites are uniform on the 1000 x 1000 grid; matrix dimensions uniform in 2-100; drum price of length j is about c x j +/- 25% (capped at 500, never proportional). Safety rules per switch cycle through three bands on tiers M/L (2.6-3.4, 3.6-4.7, 4.9-5.9), giving 9/9/8 cases (variety rule 3).

2. Planted objects and claims

Object	How the generator makes it correct by construction
Tour	Nearest-neighbour tour from site 0, then 2-opt (a segment is reversed only if the closed tour gets shorter, site 0 stays first). It is a permutation starting at site 0, tour_length is computed from it, and it is never longer than the nearest-neighbour tour (rule for L and X).
Sensor cover	The k cover nodes are chosen first (2 <= k <= n/2). Every other node gets one edge into the cover, and every further edge has at least one end in the cover, so the cover touches all edges; at least n - 1 edges are drawn.
Safety assignment	A random ON/OFF assignment is drawn first. For each random 3-switch rule, if no literal is true under the assignment, the sign of one literal is flipped, which makes the rule true. Repeated rules are rejected.
Rations	Random values 1-1000; a random 20-50% of the items is marked planted, and rations_target is defined as their sum.
Patterns, reports	Patterns 0 and 1 are cut out of the log (so they occur), pattern 5 is redrawn until it is absent; counts are found by scanning the log. report_b is report_a after k random edits and edit_bound = k, which can never be below the true distance.

Tier-S claims	Exhaustive search (generator/claims.py): every crate subset; every set of compatible windows (sorted by start, include/skip each); every way to cut the drum (all 2^{L-1} cuttings); every tour order ((n-1)! orders); node sets of size 1, 2, ... until one touches all edges. The validator recomputes all 120 claims with different methods (knapsack DP, earliest-finish greedy, weighted-interval DP, rod-cutting DP, Held-Karp, bounded search tree) and they all agree.
Variety rules	On even tier-S cases the crates are redrawn until greedy by value/kg is not optimal (holds in 13 S cases). On tier-S cases of studies 1, 3, 4, 6 the network is redrawn until some target's fewest-road routes all cost more than a cheapest route (13 cases). Separate random streams make every module of size 8 or more different (rule 4).

3. The traps (cases 46-49, tier X)

Trap	What we built	Why it is hostile	Checked by
46 Dock order (study 5)	1,603 crates. Weights sorted non-increasing, values non-decreasing (so value per kg is non-decreasing too), serials increasing: 4 of the 5 fields are monotone; only expiry is random.	Already-sorted input is the worst case of a naive quicksort pivot ($O(n^2)$) and hides bugs in sorting by value per kg; the list order itself carries no information.	V8: monotone fields counted, value/kg compared as exact fractions.
47 Best-looking crate (study 6)	17 crates, capacity 40 kg: one light decoy with value/kg above 5; 2-4 crates of 20 kg worth 100; all others are too heavy for the gap the decoy leaves and below 5 per kg.	Greedy takes the decoy, then only one 20 kg crate fits: 114 against the optimum 200 (57%).	V8: greedy value vs knapsack DP optimum.
48 Shortest looks cheapest (study 1)	1,538 locations; a chain of 601 nodes from the HQ, villages hang off chain nodes and link only inside their group, so two locations are 600 roads apart. Near the HQ: HQ-a-t (2 roads) costs 126, HQ-b-c-t (3 roads) costs 3.	A fewest-roads (BFS) route is 42 times dearer than the cheapest, and the 600-road depth makes a recursive depth-first search hundreds of calls deep.	V8: BFS from a far node; cheapest fewest-road cost vs Dijkstra.
49 Near miss (study 2)	51,016 symbols; 97.9% is one letter repeated (block length 1). Every pattern is that letter m-1 times plus a different final symbol; patterns 0-2 occur once, 3-5 never.	The first m-1 symbols match at 97.9% of all start positions, so naive matching does about L x m comparisons; KMP stays linear. Huffman sees an extremely skewed distribution.	V8: longest periodic segment (b <= 8); prefix matches per pattern.

4. What will be hard

Case study	Hardest decision 1 (data property)	Hardest decision 2 (data property)
1 Delta Flood	**Route after the waves:** 45% of locations have penalty 2 or more, so the cheapest route detours around flooded villages and differs from the fewest-roads route.	**Truck loading:** 62% of crates are food/water with similar low value per kg, giving many near-equal knapsack choices.
2 Mountain Earthquake	**Is there a route at all:** the network is almost a tree (degree at most 2.5), so one closed road often disconnects a target.	**Which alarm first:** bursts after each aftershock with mean priority above 6 mean many alarms share priority 7-9; tie-breaking by arrival decides who waits.
3 Coastal Cyclone	**Which windows to staff:** in our cyclone cases a window overlaps on average 3 to 91 others, so maximum count and maximum value pick very different sets.	**Compact transmission:** the noisy log uses at least 32 symbols with flat frequencies, so Huffman saves little over a fixed code.
4 Urban Flash Flood	**Emergency queue:** 1.5-1.8 alarms arrive per tick but one is released, so 33-44% of alarms are still waiting at the end and low priorities starve.	**Routing in a dense grid:** degree up to 4.6 gives many near-equal alternatives and the largest heaps for Dijkstra (E up to 3V).
5 Landslide Corridor	**Reachability:** the waves always cut off a target, so the engine must answer "no route" and report the reachable set.	**Cable plan:** after closures the network is a long chain (farthest location at least V/4 roads away); the tree must span only the HQ's part.
6 Outbreak Camp	**Truck loading with spoilage:** at least 55% of crates expire by hour 300, so the values of the day change quickly.	**Compression:** one beacon letter is at least 30% of the log; its Huffman code is very short, unlike in the noisy cyclone logs.